House Price Regression

Group Members

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Approach

Our group decided to work on the house price regression pre-selected challenge for our final project. Our inspiration to choose this pre-selected challenge came from the temperature regression problem that we worked on in homework 1. For the temperature regression problem that we worked on in homework 1, we worked with four different Machine Learning models. These ML models are KNN (K-Nearest-Neighbor), Naive Bayes, Lasso Linear Regression, and Ridge Linear Regression. However, for the house price regression pre-selected challenge, we implemented three of these four ML models. We implemented the KNN, Lasso Linear Regression, and Ridge Linear Regression ML models. Our purpose of using these three ML models is to provide them with the three best input parameters so that they can predict the sales price of the houses. The three best input parameters are year built, garage area, and total square feet of basement area. We found the three best input parameters in the Kaggle data that was provided to us. Here is a brief explanation of how each of these three ML models make a prediction of the sales price of the houses based on the three best input parameters that we provide:

- The KNN model computes the Euclidean Distance between the three best input parameters and all of the other training data points. It uses the k-value that we set, and it finds the K-Nearest-Neighbors in the training data. Then it takes their average house sales price, and makes a prediction based on the information that it gained.
- The Lasso Linear Regression model uses L1 regularization, and it predicts the house sales price by calculating the weighted sum of the three best input parameters that we provide it (the Lasso Linear Regression model also uses the coefficients that it learned in the linear model).
- The way that the Ridge Linear Regression model predicts the house sales price is very similar to how the Lasso Linear Regression model predicts the house sales price. The important difference to understand here is that the Ridge Linear Regression model uses L2 regularization instead of L1 regularization.

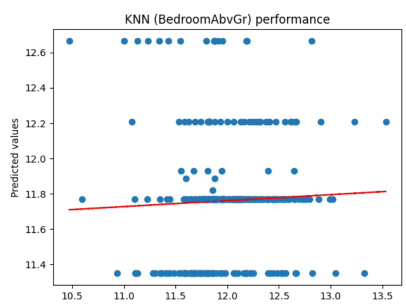
Moreover, we trained these three ML models based on the three best parameters and calculated the RMSE (Root Mean Squared Error) values, and the MAE (Mean Absolute Error) values. These values are predicted between the logarithm of the predicted value, and the logarithm of the observed sales price. We created a table that contains the RMSE values, and MAE values (which we will discuss in one of the next sections). We did this to have a clear understanding of the results that we obtained. Furthermore, we created a graph for each of the three ML models between the predicted values and the true values. These three graphs also contain a line of best fit to see how closely the predicted values matched the true values. We did look for a way to clean the house price regression data found on Kaggle, but after thoroughly looking into the data, we realized that there was no particular reason to clean the data since the Sales Prices and parameters we chose had no values with None or Zero. So far, we discussed the approach of doing our work on the house price regression pre-selected challenge for our final project. The rest of the paper is organized as follows. We will now discuss our implementation process. Then we will discuss all the experiments that we have conducted. Lastly, we will discuss our conclusions, and what we could have done differently.

Implementation

1. Tensorflow: Tensorflow is an open source machine learning library made by Google which allows us to build, train, and deploy machine learning algorithms and models. It has a multitude of features which allow it to build models in a scalable and efficient manner. Tensorflow is also used to build deep neural networks.
2. Pandas is a data manipulation and analysis library for python which allows us to take structured data in CSV or excel format files and manipulate the data to fit our needs. In this project we used Pandas to help us use specific columns such as "TotalBsmtSF, GarageArea, and Year Built " that we needed to analyze for our prediction models.
3. Scikit-Learn: Scikit-Learn is a popular machine learning library which is used mainly to build and evaluate ML models. Using this model we were able to implement Lasso, Ridge, and the KNeighborsClassifier. We also utilized sklearn.metrics and sklearn.linearModel to implement Median Absolute Error and Lasso respectively.
4. Matplotlib: Matplotlib is a popular data visualization library that we used to build high level statistical graphics such as the scatter plots and linear regression graphs.

Experiments

We performed experiments on our models using the following parameters: LotArea (Area of lot), YearBuilt (Original construction date), TotalBsmtSF (Total square feet of basement area), and GarageArea (Size of garage in square feet). We initially tried to use the Bedrooms as a parameter, but that produced a graph that wasn't linear enough, and instead was split into several horizontal lines as shown on the right. We thus settled on the 4 parameters above.

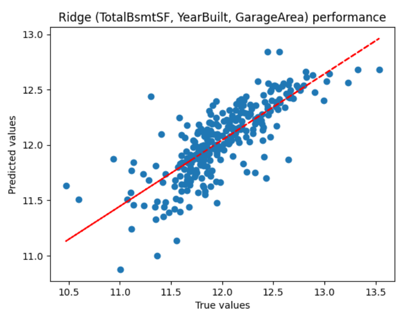


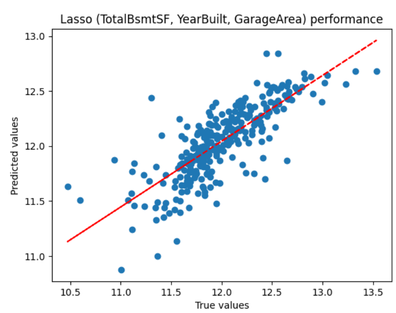
For the KNN model, we chose an N value of 3 since it produced the best RMSE and MAE values with a linear diagonal line of best fit.

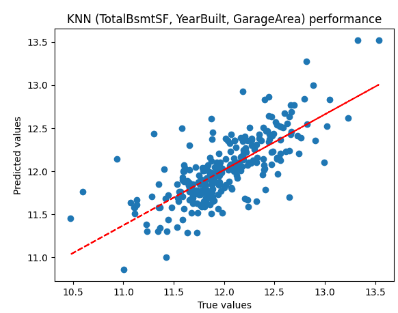
We ran the three models using one parameter at a time to understand which parameters were the best predictors for Sales Price. Below is a table of the Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). GarageArea produced the lowest error values, followed by TotalBsmtSF and YearBuilt. LotArea seemed to deviate too far from the RMSE and MAE values compared to the other three parameters, so we chose to ignore it in training our models.

MODEL	LotArea	YearBuilt	TotalBsmtSF	GarageArea
Lasso	rmse 0.4289 mae 0.2861	rmse 0.3595 mae 0.2412	rmse 0.3421 mae 0.2444	rmse 0.3461 mae 0.1629
Ridge	rmse 0.4289 mae 0.2861	rmse 0.3595 mae 0.2412	rmse 0.3421 mae 0.2444	rmse 0.3461 mae 0.1629
KNN - 3	rmse 0.5045 mae 0.3212	rmse 0.4115 mae 0.2128	rmse 0.3942 mae 0.2185	rmse 0.3838 mae 0.1915

Finally, we trained our 3 models based on GarageArea, YearBuilt, and TotalBsmtSF and plotted the Predicted versus True values for the Sales price for each model below. We plotted a line of best fit to visualize how close the predicted values were to the True values. The closer the dots are to the line of best fit, the better the model is at predicting.







Based on the graphs above, it looks like Ridge and Lasso produce similar but better predictions than KNN which has a slightly more spread out plot. In future experiments, we can add weights to different features that are better predictors for sales price, and test out different alpha values for the regression models.

Discussion:

Based on the given data we can see that the Lasso and Ridge Models have the best performance. We can see that they have the lowest Root Mean Squared Error (RMSE) and the Mean Absolute Error (MAE) compared to KNN. On the other hand we see that the KNN model with K=3 has performed worse. This means that the Lasso and Ridge Models are significantly better at predicting house prices in comparison to the other models given our current features. We also need to remember that the conclusions we have drawn from the data set only apply when we have the same features and same size of dataset.

Because this data is not definitive, in the future we can use different algorithms and a different set of parameters to improve the performance of our model. In the future we can also compare and contrast different features and select those which may fit our needs better to reduce the potential impact of multicollinearity on the model's performance.

Citations/Acknowledgements

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